Assignment #2: 位运算、前缀和、树状数组、归并排序 & 状态压缩

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1. 题目

E868.二进制间距

bit manipulation, https://leetcode.cn/problems/binary-gap/

> 主要是练习面向对象编程写法，这样力扣题目，笔试都没有问题了。机考时候，不是必须OOP，能AC就可以。

思路：

代码：

```
class Solution:
    def binaryGap(self, n: int) -> int:
        s = bin(n)[2:]
        gap = 0
        begin = s.index("1")
        end = begin
        while end < len(s):
            if begin == end:
                end += 1
            elif s[end] != "1":
                end += 1
            elif s[end] == "1":
                gap = max(gap,end-begin)
                begin = end
        return gap
```

Status	Language	Runtime
1 Accepted Mar 06, 2026	Python3	0 ms

Python3 Auto

```python
            if begin == end:
                end += 1
            elif s[end] != "1":
                end += 1
            elif s[end] == "1":
                gap = max(gap,end-begin)
                begin = end
        return gap
```

Saved Ln 15, C

Testcase | Test Result

You must run your code first

M304.二维区域和检索 - 矩阵不可变

prefix sum, https://leetcode.cn/problems/range-sum-query-2d-immutable/

思路:

代码:

```python
class NumMatrix:
    def __init__(self, matrix):
        self.matrix = matrix
        n = len(matrix)
        m = len(matrix[0])
        self.dp = [[0 for _ in range(m+1)] for __ in range(n+1)]
        for i in range(1,n+1):
            for j in range(1,m+1):
                self.dp[i][j] = self.dp[i-1][j]+self.dp[i][j-1]-self.dp[i-1][j-1]+self.
        print(self.dp)
    def sumRegion(self, row1: int, col1: int, row2: int, col2: int) -> int:
        row1 += 1
        row2 += 1
        col1 += 1
        col2 += 1
        return self.dp[row2][col2]-self.dp[row2][col1-1]-self.dp[row1-1][col2]+self.dp[
```

Accepted 22 / 22 testcases passed

cZ88PGJiHH submitted at Mar 11, 2026 13:27

Runtime

145 ms | Beats 25.00%

Memory

34.98 MB | Beats 5.54%

30%

```
[j-1]+self.matrix[i-1][j-1]
        print(self.dp)
    def sumRegion(self, row1: int, col1 int, row2: int, col2: int) -> int:
        row1 += 1
        row2 += 1
        col1 += 1
        col2 += 1
        return self.dp[row2][col2]-self dp[row2][col1-1]-self.dp[row1-1][col2] +self.dp[row1-1][col1-1]
```

Saved Ln 16, Col

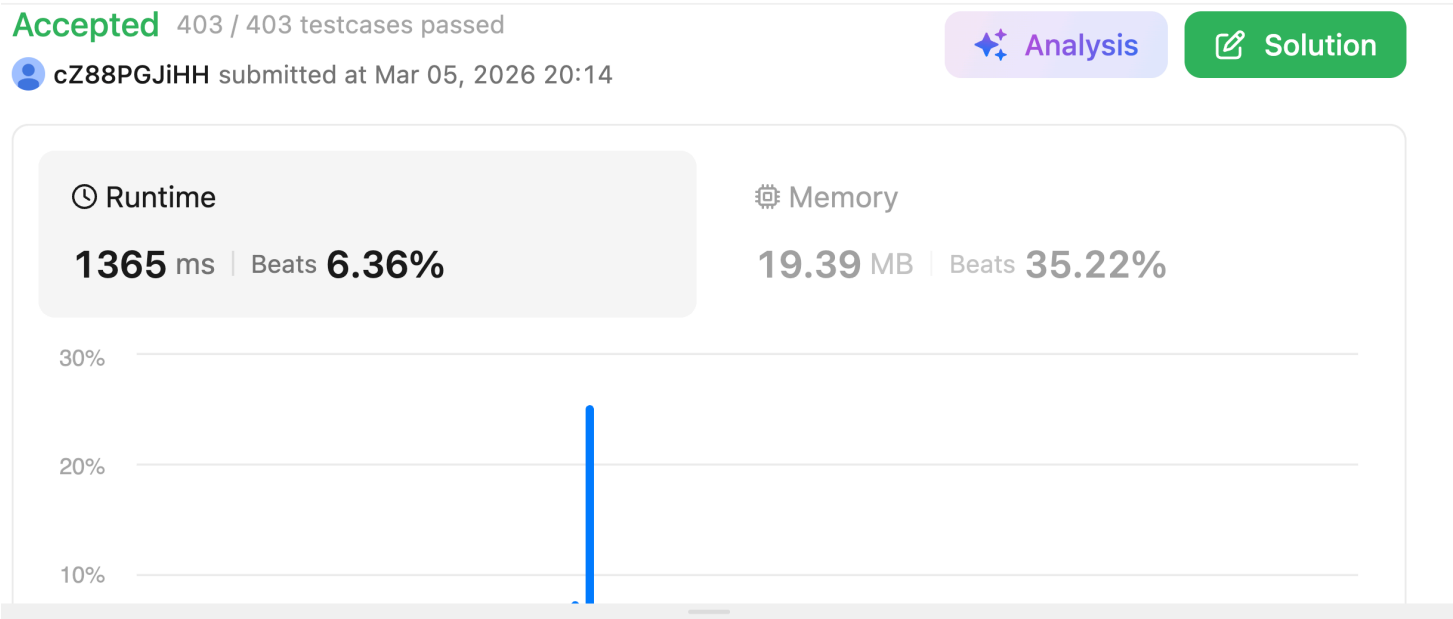
M1680.连接连续二进制数字

bit manipulation, https://leetcode.cn/problems/concatenation-of-consecutive-binary-numbers/

思路：

代码：

```
class Solution:
    def concatenatedBinary(self, n: int) -> int:
        zhi = 0
        ermi = 1
        for t in range(n,0,-1):
            zhi += ermi*t
            zhi %= (10**9+7)
            ermi *= ((2**(len(bin(t))-2))%(10**9+7))
            ermi %= (10**9+7)
        return zhi
```



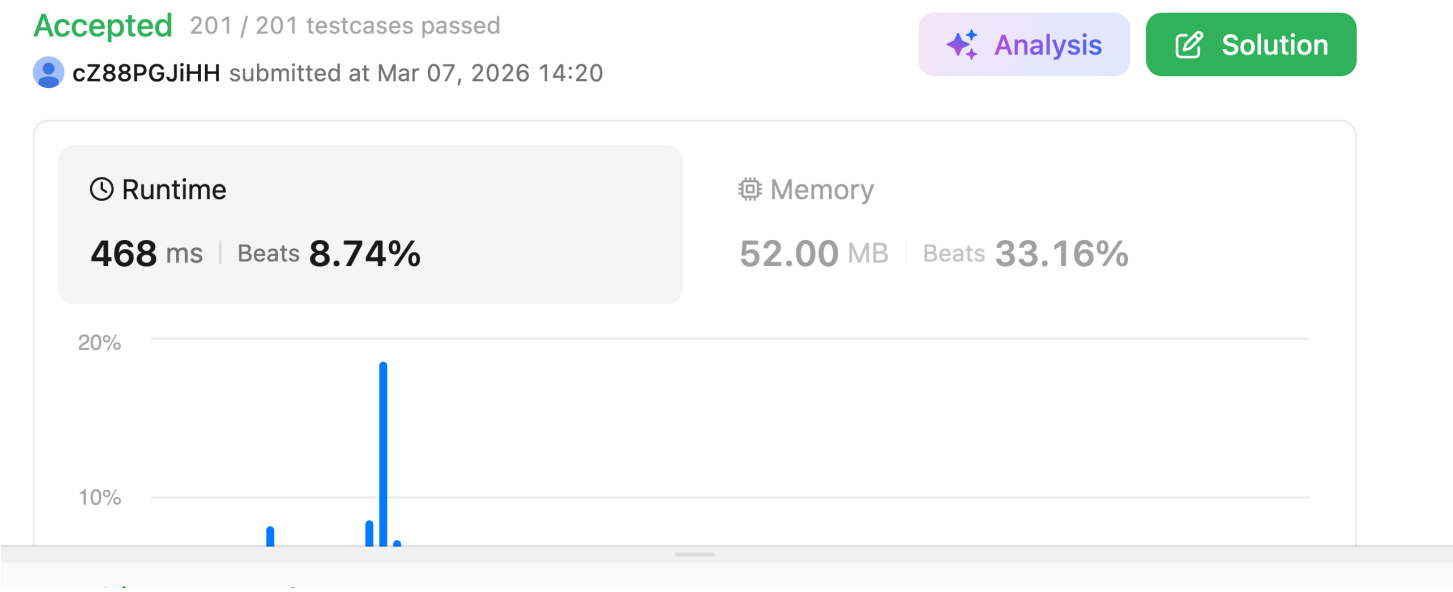
M1461.检查一个字符串是否包含所有长度为 K 的二进制子串

bit manipulation, https://leetcode.cn/problems/check-if-a-string-contains-all-binary-codes-of-size-k/

思路：

代码：

```
class Solution:
    def hasAllCodes(self, s: str, k: int) -> bool:
        if len(s) < k:
            return False
        storage = set()
        curstr = [s[j] for j in range(k-1)]
        for i in range(len(s)-k+1):
            curstr.append(s[i+k-1])
            storage.add("".join(curstr))
            curstr.pop(0)
        if len(storage) == (1<<k):
            return True
        return False
```



M30178:数字华容道（Easy Version）

merge sort, binary indexed tree, http://cs101.openjudge.cn/practice/30178/

思路：

代码：

```python
n = int(input())
sequence = []
for i in range(n):
    mylist = list(map(int,input().split()))
    if 0 in mylist:
        s = n-i+1
        mylist.remove(0)
    sequence.extend(mylist)
shu = [0 for _ in range(n**2)]
def update(i):
    while i < len(shu):
        shu[i] += 1
        i += (i&(-i))
def query(t):
    total = 0
    while t > 0:
        total += shu[t]
        t -= (t&(-t))
    return total
count = 0
for num in sequence:
    num = n**2-num
    count += (query(num)%2)
    count %= 2
    update(num)
if n % 2 == 1:
    if not count:
        print("yes")
    else:
        print("no")
else:
    if (count + s) % 2 == 0:
        print("yes")
    else:
        print("no")
```

#51954610提交状态

View Submit Statistics

状态: Accepted

基本信息

#: 51954610
题目: 30178
提交人: 25n2500010774(sleepy(25n25
内存: 58152kB
时间: 4919ms
语言: Python3
提交时间: 2026-03-12 22:26

Source Code

```
n = int(input())
sequence = []
for i in range(n):
    mylist = list(map(int,input().split()))
    if 0 in mylist:
        s = n-i+1
        mylist.remove(0)
    sequence.extend(mylist)
shu = [0 for _ in range(n**2)]
def update(i):
    while i < len(shu):
        shu[i] += 1
        i += (i&(-i))
```

T30201: 旅行售货商问题

bitmask dp, http://cs101.openjudge.cn/practice/30201/

思路:

代码:

```python
from collections import deque
n = int(input())
cost = []
for _ in range(n):
    cost.append(list(map(int,input().split())))
k = (1<<n)
matrix = [[float("inf") for _ in range(n)] for __ in range(k)]
matrix[1][0] = 0
minimum = float("inf")
for visit in range(2,k):
    for cur in range(1,n):
        if visit & 1 == 0:
            continue
        if (1<<cur)|visit != visit:
            continue
        visitnew = visit - (1<<cur)
        for i in range(n):
            if (1<<i)|visitnew != visitnew:
                continue
            a = matrix[visitnew][i]+cost[cur][i]
            if a >= matrix[visit][cur]:
                continue
            matrix[visit][cur] = a
for i in range(1,n):
    minimum = min(minimum,cost[i][0]+matrix[k-1][i])
print(minimum)
```

#51937095提交状态

View Submit Statistics Clarify

状态: Accepted

Source Code

```python
from collections import deque
n = int(input())
cost = []
for _ in range(n):
    cost.append(list(map(int,input().split())))
k = (1<<n)
matrix = [[float("inf") for _ in range(n)] for __ in range(k)]
matrix[1][0] = 0
minimum = float("inf")
for visit in range(2,k):
    for cur in range(1,n):
        if visit & 1 == 0:
            continue
        if (1<<cur)|visit != visit:
```

基本信息

#: 51937095
题目: 30201
提交人: 25n2500010774(sleepy(25n2500010774)
内存: 99860kB
时间: 837ms
语言: PyPy3
提交时间: 2026-03-11 14:33:09

2. 学习总结和个人收获

月考1小时ak，还是很轻松的，说明上学期学的东西还挺扎实，必须得感谢闫老师给我打下的基础
数字华容道没有想到用逆序对www，以前搞竞赛的时候组合还是学的太差了，最后是被同学提示后意识到逆序对这事后遂做出
由于这周比较忙，没有刷其他的题，只是把讲义做了一下（做讲义才发现原来华容道的tag就是暗示逆序对（www没看到tag）